

DUPLICATE ORIGINAL

Contract.4600000582
RPO No.16

**Release Purchase Order for Master Research Agreement (No. 4600000582)
Release Purchase Order (RPO) No.16**

Upon execution by the parties below, the Research Project specified herein shall be awarded and performed in accordance with the Master Research Agreement ("Agreement", No. 4600000582), which shall be incorporated herein in its entirety. The project specifications shall include the following:

a. The Research Project title:

Assessment of Low-Carbon Development Technologies in the iron and steel industry:

b. A description of the purpose of the project:

The global iron and steel industry faces a significant challenge to achieve carbon reduction targets due to its high carbon emissions intensity. This industry urgently needs to explore innovative low-carbon emission reduction technologies and feasible strategies in order to address climate change concerns. The proposed study will focus on assessing the potential technology pathways to reduce carbon emissions in the iron and steel industry in China, and will help gain critical insight into the potential impact of low-carbon energy technologies in hard-to-abate sectors of the economy. The project will primarily include a detailed assessment of low-carbon energy technologies in the iron and steel industry, and a regional low-carbon emissions pathways model to depict the technology deployment and interregional industrial chain structure, including the transportation of raw materials and finished products across different regions in China.

c. A detailed scope of work and implementation plan to include the contribution of both University and AAJ, as applicable, and all milestones and deliverable for each phase of the project (Attachment):

Scope of Work is shown in Attachment I of RPO No.16.

d. Duration of the project: Start Date: October 28, 2023, End Date: April 27, 2025

e. A list of any funding requirements for this Research Project (Attachment):

None.

f. A proposal for who should act as Principal Investigator in the project:

Tsinghua University Principal Investigator of this CONTRACT shall be:

Contact Name: Prof. Zheng Li
Telephone: +86 010-62795736
E-Mail: lz-dte@tsinghua.edu.cn

Co-PI : Dr. Tianduo Peng
Telephone : +86 010-62786891
E-Mail : ptd@mail.tsinghua.edu.cn

g. The Saudi Aramco project representative:

Saudi Aramco Representative for the purposes of administering this CONTRACT shall be:

Contact Name: Daniel De Castro Gomez
Telephone: +966 13 8808457
E-Mail: daniel.decastrogomez@aramco.com

Or another duly appointed delegate.

Saudi Aramco Representative must be copied on all reports and electronic communication between Tsinghua University and SAUDI ARAMCO.

h. A brief description of any confidential information that will be shared:

The confidential information is defined in article 8 of the Agreement.

i. A description of any background IP that will be used as part of the proposed project:

None.

j. A description of any pre-existing work that is being done by University in the proposed area of research, to include informing AAJ of any third Parties that are or that may become involved in the project:

None

k. Intellectual property ownership provisions, if different from the terms specified in this Agreement.

The Intellectual Property provisions remain the same as in the Master Research Agreement.

I. Financial Accounting:

Amount of Funding from AAJ:

- i. Amount for this action\$300,000.00

m. Invoicing Frequency:

- i. ☐ Quarterly
ii. ☒ Other (refer to Attachment 1 of RPO No. 16)

n. Technical Reporting Requirements:

- i. ☒ Progress Reports;
ii. ☐ Quarterly Meetings
iii. ☐ Final Report (90 days post term)

o. Other terms and conditions:

COMPANY shall acknowledge receipt of each of the above-mentioned deliverables.

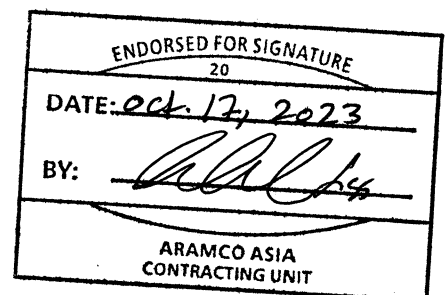
The duly authorized representative of the University hereby agree to accept the terms of this RPO as indicated by their signatures below.

Tsinghua University

Aramco Asia Japan K.K.

Signature: _____
Acknowledged and accepted by
Principal Investigator:

Signature: _____
Name: Abdullah Jastaniah
Title: Representative Director



Contract Release Purchase Order (CRPO) No. 16

Scope of Work

Assessment of Low-Carbon Development Technologies in the Iron and Steel Industry

1. Background

With the increasing global concern about climate change and environmental issues, carbon emissions in the industrial sector are becoming more prominent. The iron and steel industry, as one of the energy-intensive industries, poses a significant challenge to achieving global carbon reduction goals due to its high carbon emissions. To address climate change, this industry urgently needs to explore innovative low-carbon emission reduction technologies and feasible strategies to achieve sustainable development. Since the beginning of this century, China's iron and steel industry has seen a significant increase in production and emissions, with a high dependence on coal in its energy mix. In 2020, China's iron and steel industry emitted approximately 1.9 billion tons of carbon dioxide, accounting for about sixty percent of global steel production and emissions. In the context of carbon neutrality, it is essential for China's steel industry to develop low-carbon and deep emission reduction strategies.

The steel industry boasts a plethora of low-carbon technologies, ranging from the utilization of environmentally friendly fuels like natural gas and biomass to carbon capture and storage (CCS). Equally significant is the role of hydrogen in the steel industry's decarbonization efforts. By using hydrogen as a reducing agent in the direct reduction of iron ore or as a clean fuel in electric arc furnaces, steelmakers can significantly cut their carbon footprint. Understanding the emissions reduction potential and economic viability of these diverse technological pathways, as well as discerning the evolving trends within each, remains a critical task. This is especially important in the context of global efforts to combat climate change and achieve net-zero emissions.

The impending energy transition could usher in profound transformations within the steel sector. There's a likelihood that the industry may gravitate towards regions abundant in mineral resources and fortified by clean energy. In May 2023, Saudi Aramco, Baoshan Iron & Steel Co., and the Saudi Public Investment Fund inked a shareholder agreement to build a state-of-the-art, fully integrated steel factory in the Kingdom of Saudi Arabia. This strategic move underscores the steel industry's commitment to explore new avenues for sustainable steel production. We aspire that our research can serve as a valuable reference for the low-carbon development of steel production in Saudi Arabia.

2. Research Objectives and Content Design:

This study will focus on the development of low-carbon emission reduction in the iron and steel industry and primarily cover the following two important research areas:

Task 1: Assessment of Low-Carbon Development Technologies in the iron and steel industry:

The main goal of this topic is to evaluate the feasibility, economic benefits, and potential cost reduction of low-carbon development technologies in the iron and steel industry. It will comprehensively analyze the future

development trends of low-carbon technologies in the iron and steel industry, thereby providing a basis for technology selection for industrial upgrading and scientific support for setting emission reduction targets and policy decisions. Specific objectives include:

- Evaluate the application potential of various low-carbon technologies: Analyze various low-carbon technologies such as natural gas, biomass, carbon capture and storage (CCS), and hydrogen and assess their applicability and potential in the steel production process.
- Explore the carbon emission reduction potential of low-carbon technologies: Evaluate the carbon emission reduction potential of different low-carbon technologies in steel production and analyze their impact on carbon emissions at different time stages of the production process.
- Analyze the economic benefits of low-carbon technologies: Assess the cost of different low-carbon technologies in the iron and steel industry.
- Predict the future development trends of low-carbon technologies: based on technological innovation and market demand, forecast the future development trends of different low-carbon technologies and the potential for cost reduction.
- Qualitative sum up the impact of technology dissemination: From environmental, economic, and social perspectives.

Task 2: Study on the Evolution of the Industrial Layout of the iron and steel industry:

Based on the results in Task 1, assess the industrial layout evolution of China's steel industry driven by the technological revolution. Develop a regional low-carbon emission reduction pathway model to depict the regional technology development and interregional industrial chain layout, including the transport of raw materials and products across regions. Characterize the temporal and spatial attributes of technological development. This analysis will yield deep decarbonization pathways at the provincial level, considering demand trends and regional resource endowments from the optimal perspective of the industrial chain. Based on this, propose reasonable and feasible layout planning methods to guide the design of provincial transition pathways and industrial chain layout policies. Specific objectives include:

- Analyze the characteristics of the industrial chain: Based on the material and energy flow analysis of various steel production technologies, describe the differences in the industrial chain and calculate the total supply chain cost of steel from a supply chain perspective.
- Establish a multi-regional low-carbon development pathway model: Combining top-down demand forecasting and bottom-up technology selection, establish a planning model (2020~2060) at the provincial level and with the annual resolution. Take China's steel industry as a case study, considering the differences in regional resource endowments. The optimization of the whole industrial chain is included. Generate provincial low-carbon development pathways for China's steel industry based on the comprehensive assessment of the future development of low-carbon technologies. Depict the interregional transfer of raw materials and products.

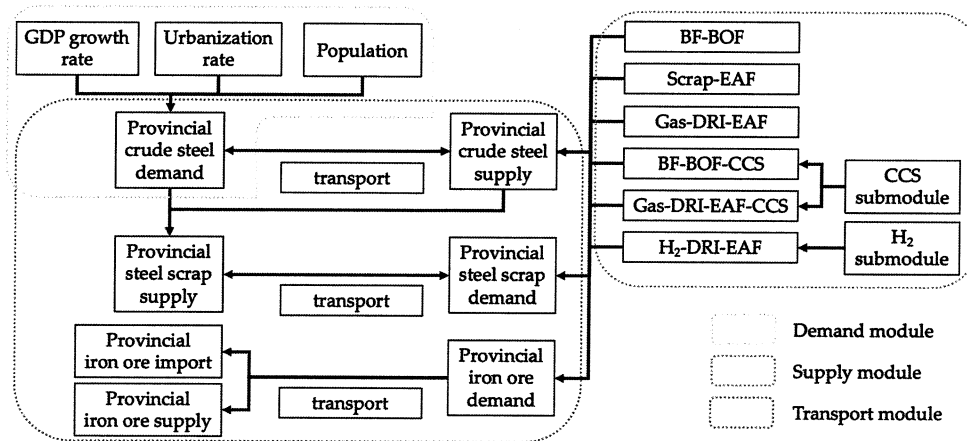


Fig. Model Framework

- Evaluate the temporal and spatial distribution of low-carbon technologies: reveal the temporal and spatial distribution patterns of low-carbon technologies and compare the characteristics of technology structures and breakthrough technology applications in various provinces.
- Characterize the evolution of industrial layout: Analyze the impact of the technological revolution on the industrial layout of China's steel industry and quantitatively depict the spatio-temporal evolution under the scenario of deep emission reduction.
- Examine the impact and strategies of interprovincial capacity transfer: Analyze the impact of interprovincial capacity transfer on different regions and its implications for future supply chain design, and explore the feasibility, challenges, and strategic planning of interprovincial capacity transfer.

3. Research Arrangements:

- 3.1 Literature Review: Summarize the current state of technology development and industry trends, and gain an understanding of the characteristics of the steel industry both domestically and internationally.
- 3.2 Policy Review: Develop a toolbox of emission reduction policies for China's steel industry, interpret the emphasis and policy changes over different time periods, and summarize the characteristics of the development and transition of China's steel industry.
- 3.3 Professional Exchange: Organize a series of "Steel +" salons under the project, such as "Steel + Hydrogen," "Steel + CCUS," etc. Invite institutions and experts with extensive experience in the fields of energy, steel, hydrogen, CCUS, and more to exchange ideas. Hold regular project discussions in conjunction with project progress.
- 3.4 Data Collection: Establish a spatiotemporal emission reduction database. Collect macro-level and industry-specific technical data, including the distribution of China's crude steel market, future demand projections, production technology development trends, and the driving factors behind industrial layout evolution.
- 3.5 Field Research: Conduct on-site investigations of transition practices in iron and steel enterprises such as Baowu or HBISGROUP. Explore the current status, the challenges of emission reduction, and ongoing practices such as DRI of steel companies. Analyze the technological preferences of different regions and enterprises based on their roadmaps.

3.6 Model Development: Build a low-carbon development pathway model for the steel industry based on technology assessments. Utilize high-performance experimental equipment and computing servers to support the complex models in the research process.

4. Deliverables:

The project is scheduled for a duration of 18 months with an anticipated project start on October 28, 2023.

Phase 1: Planned Technical Progress (Oct 28, 2023 – July 27, 2024)

Task 1. Assessment of Low-Carbon Development Technologies in the iron and steel industry:

- 1) Literature Review and Policy Analysis: Collect international and domestic research, application cases, and policies related to low-carbon technologies.
- 2) Technology Assessment: Evaluate the applicability of low-carbon technologies, considering technical feasibility, performance parameters, and carbon emission reduction potential. Establish cost models and carbon emission accounting models to calculate the cost composition and environmental impact of different production technologies.
- 3) Future Trends Forecast: Predict the future cost trends of low-carbon technologies. Comprehensively assess technology promotion by considering technology assessments, cost-benefit analysis, and environmental impact.

Milestones and Deliverables

- 4.1.1 Report on literature and policy review, and future cost and carbon reduction trends of low-carbon technologies analysis.
- 4.1.2 Cost and carbon emission accounting models
- 4.1.3 Associated model historical and forecast data which could include CAPEX, OPEX, technology costs curves for conventional and low-carbon technologies, carbon emissions, fuel prices, macroeconomic and policy indicators amongst others.
- 4.1.4 Submit a paper to a reputable journal.

Phase 2: Planned Technical Progress (July 28, 2024 – April 24, 2025)

Task 2. Study on the Evolution of the Industrial Layout of the iron and steel industry:

- 1) Literature Review and Data Collection: Collect provincial-level production data and summarize the characteristics of the industrial layout of China's steel industry. Analyze the driving factors of the industrial layout evolution in the domestic and international steel industry development process.
- 2) Model Development: Build a multi-regional low-carbon development pathway model including the entire industrial chain, based on various low-carbon technologies in Task 1.

- 3) Scenario Building, Results Analysis, and Summary: Create multiple scenarios and summarize research results, including regional technological development, inter-provincial capacity transfers, raw material and product transport. Summarize the patterns of future industrial layout evolution.

Milestones and Deliverables

- 4.2.1 Multi-regional low-carbon development pathway model.
- 4.2.2 Associated model historical and forecast data which could include CAPEX, OPEX, technology costs curves for conventional and low-carbon technologies, carbon emissions, fuel prices, macroeconomic and policy indicators amongst others.
- 4.2.3 Comprehensive research report on provincial-level production and characteristics of the industrial layout, scenarios and results on the regional technology development, and patterns of future industrial layout.
- 4.2.4 Submit a paper to a reputable journal.

5. Payment Schedule

Start Date: October 28, 2023 End Date: April 27, 2025 Duration: 18 months

The percent of lump sum for the payment schedule shall be according to the following:

Payment No.	Delivery Schedule	Milestone	% of Lump Sum	Amount (\$)
1	Upon execution of the RPO No. 16	Execution of the RPO No. 16 by all parties	50%	\$150,000.00
2	9 months after RPO execution	Acceptance by Aramco of reports and any other deliverables submitted by Tsinghua University in accordance with Milestones 4.1.1, 4.1.2, and 4.1.3	25%	\$75,000.00
3	18 months after RPO execution	Acceptance by Aramco of reports and any other deliverables submitted by Tsinghua University in accordance with Milestones 4.2.1, 4.2.2, and 4.2.3	25%	\$75,000.00
Total			100%	\$300,000.00

** No payment will be due until thirty (30) days after completion of respective milestones subject to the acceptance of Aramco.*

** Any payment Amount has included all applicable taxes. In case VAT or related tax is applicable in concerned countries, this should be reported to Aramco prior to submission of any invoice. In case withholding tax is applicable in concerned countries, the amount of tax to be withheld shall be deducted from any Amount payable by Aramco before transferring payment.*

6. CONTACT PERSONS

Company Representative for the purposes of administering this work shall be:

Contact Name: Daniel De Castro Gomez

Telephone: +966 13 8808457
Email: daniel.decastrogomez@aramco.com

Tsinghua University Principal Investigator of this CONTRACT shall be:

Contact Name: Prof. Zheng Li
Telephone: +86 010-62795736
E-Mail : lz-dte@tsinghua.edu.cn

Co-PI: Dr. Tianduo Peng

Telephone: +86 010-62780968 +86 13671058362
E-Mail: ptd@mail.tsinghua.edu.cn

The Company Representative must be copied on all reports and electronic communication between Tsinghua University and Aramco. Tsinghua University shall identify Contact Person to be available at all times during performance of WORK to respond to Company Representative.

7. MANAGEMENT PLAN

WORK shall be carried out by Tsinghua University. WORK shall be directed by the Tsinghua University Representative and communicated to the Company Representative as per the deliverables listed above and issues relating to any technical and management problems shall be solved through discussions by both parties.